

AWS EC2 and EBS Volume Attachment Report

Step-by-step practical report for launching an EC2 Ubuntu instance, creating an EBS volume, attaching the volume to EC2, and verifying storage mounting through Linux terminal commands.

AWS Region: Asia Pacific (Mumbai) | Operating System: Ubuntu 24.04 LTS

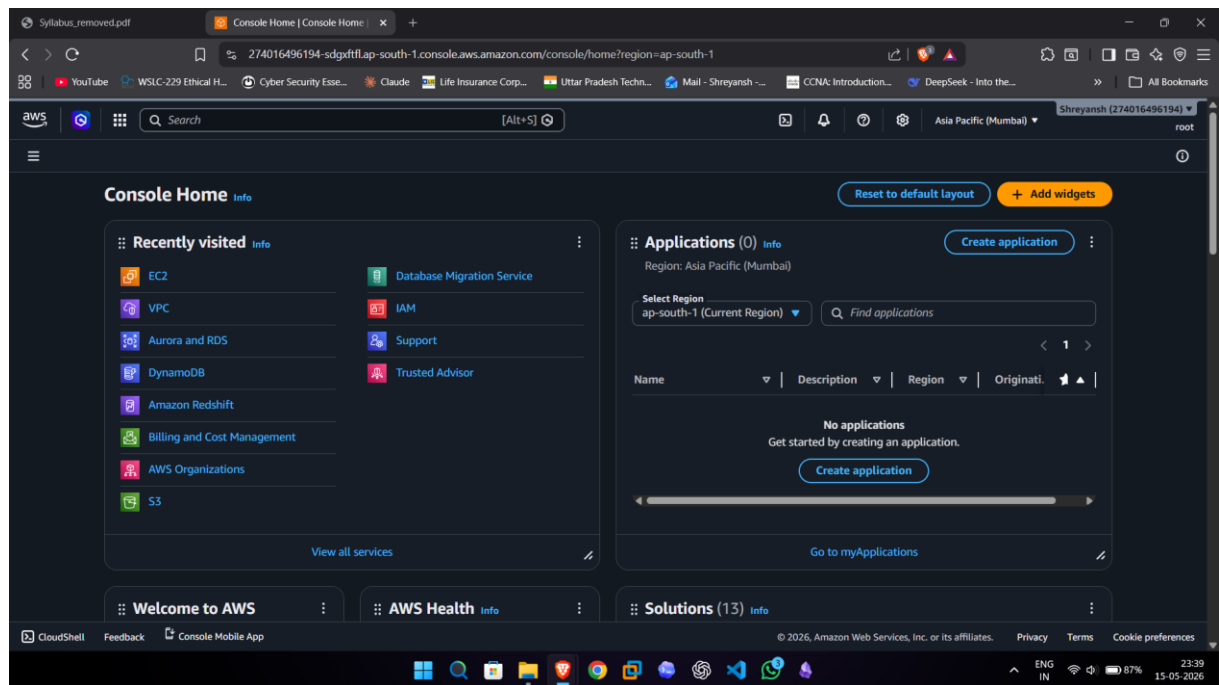
Practical Overview

This practical demonstrates how to:

- Launch an EC2 Ubuntu instance
- Create and configure an EBS volume
- Attach the EBS volume to the EC2 instance
- Connect using SSH from Windows Terminal
- Format and mount the EBS volume in Ubuntu Linux
- Verify data persistence using Linux commands

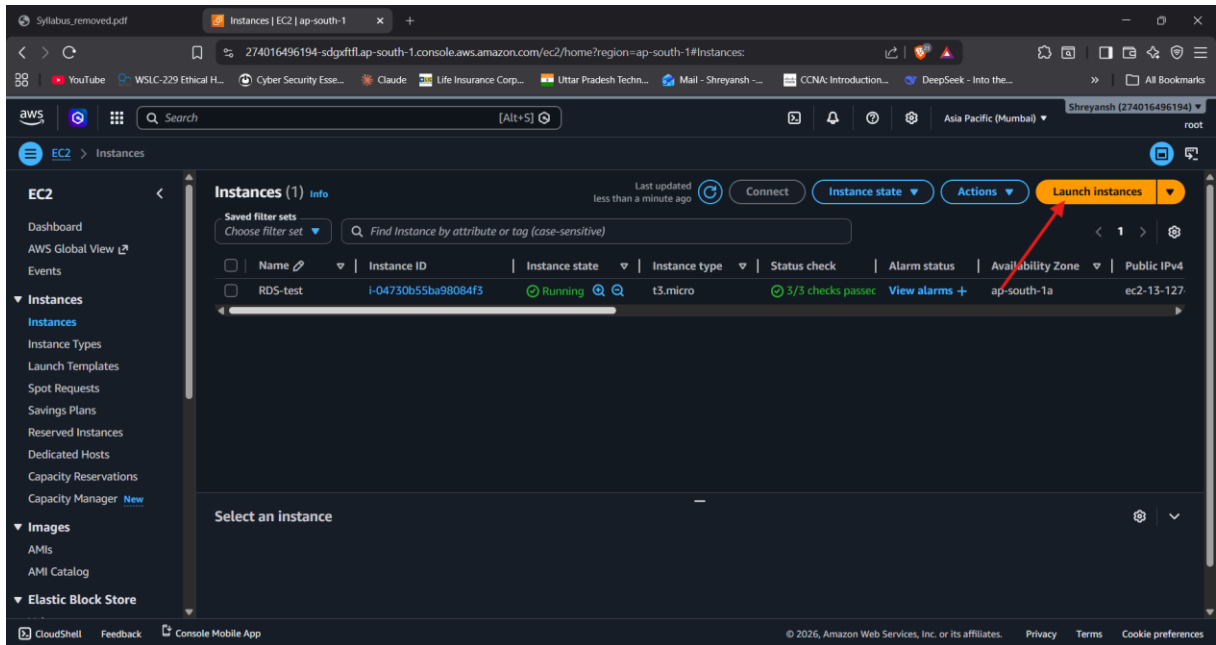
Step 1: Open AWS Console Home

AWS Console Home page opened successfully in the Mumbai region.



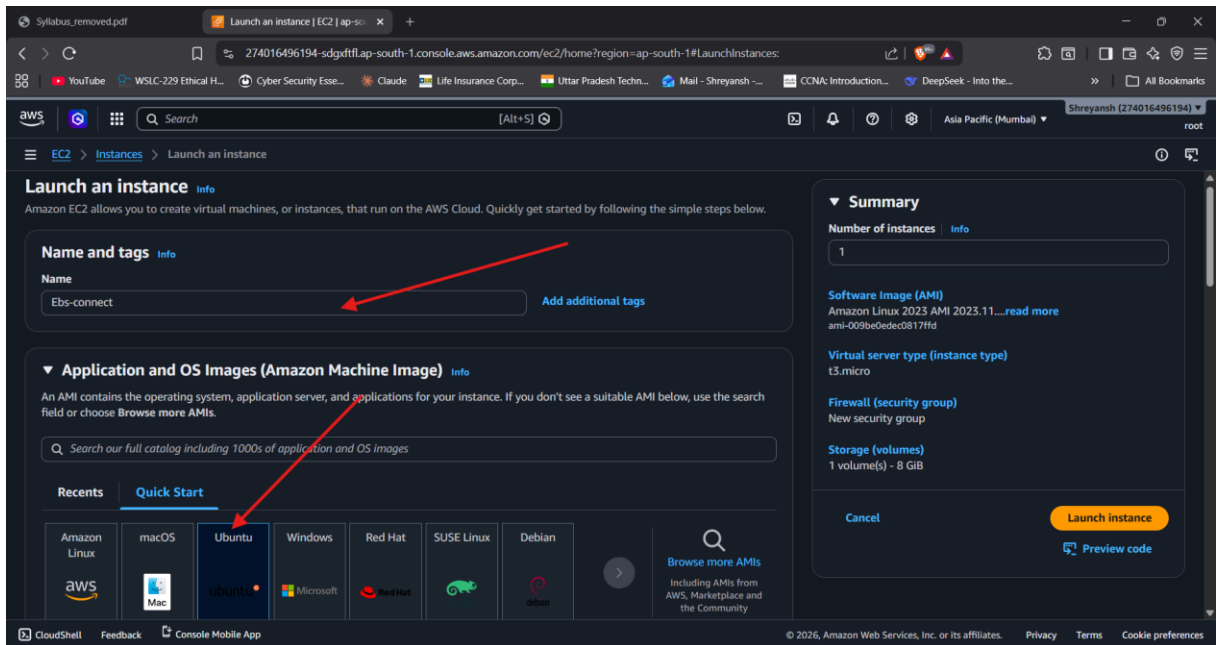
Step 2: Open EC2 Instances Dashboard

The EC2 Instances page shows the currently running instance.



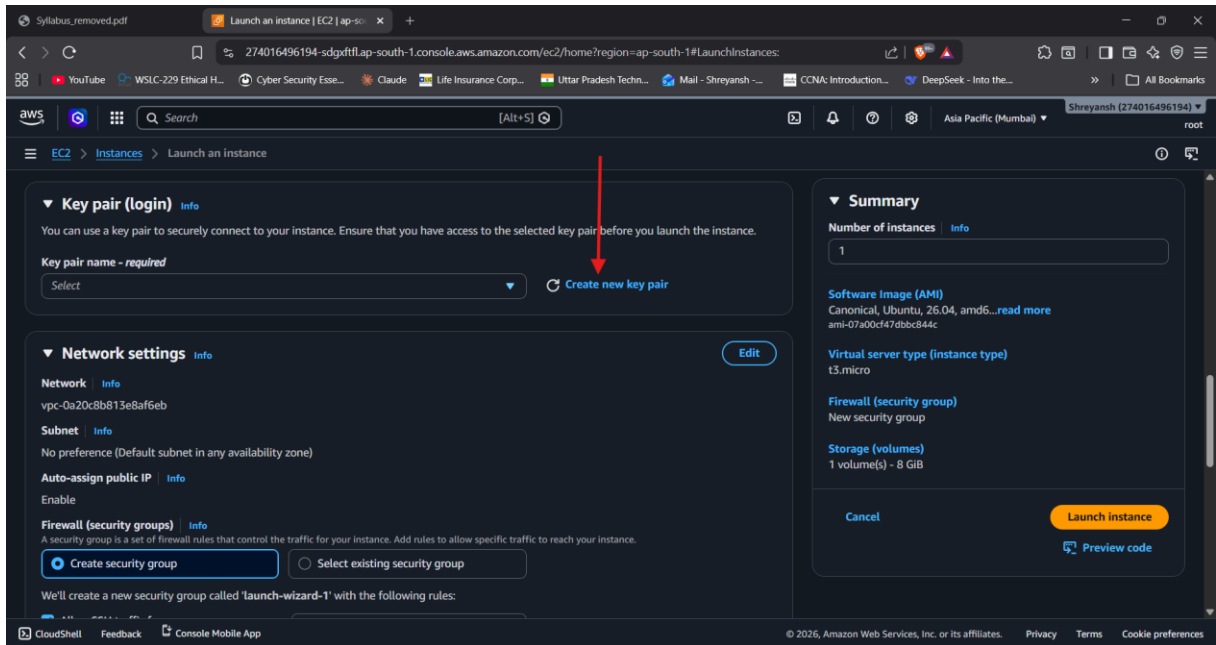
Step 3: Launch a New EC2 Instance

A new Ubuntu EC2 instance configuration page is opened.



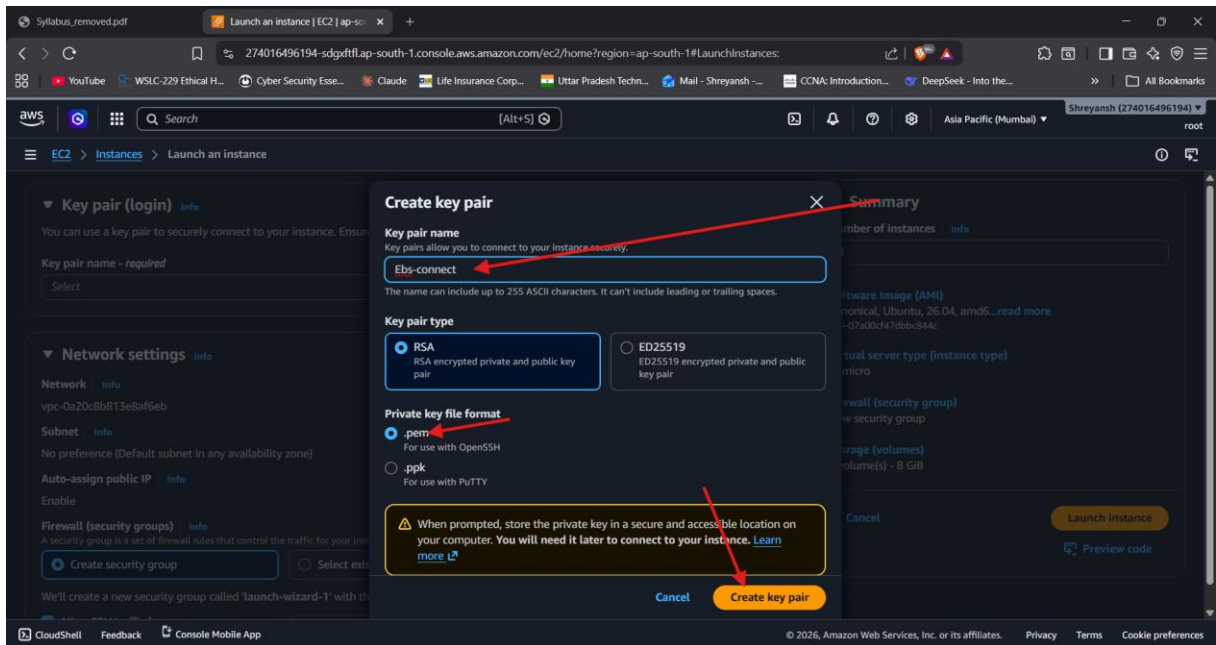
Step 4: Configure Key Pair

A new RSA key pair with .pem format is created for SSH authentication.



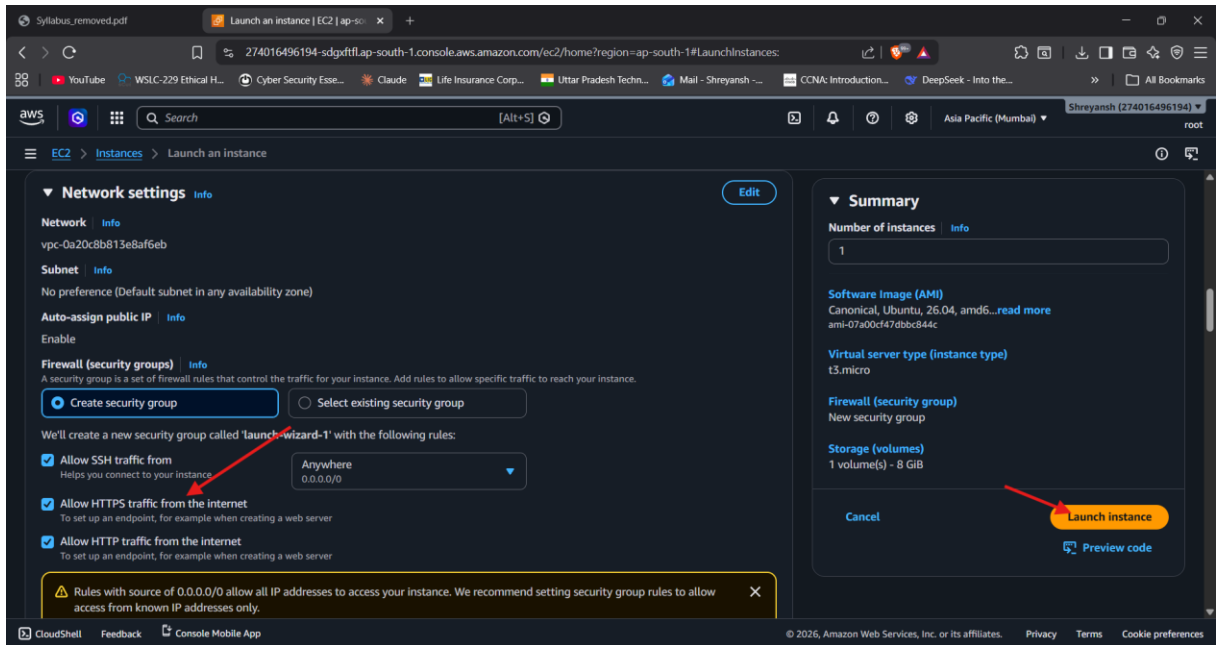
Step 5: Configure Security Group and Launch Instance

SSH, HTTP, and HTTPS inbound rules are enabled before launching the instance.



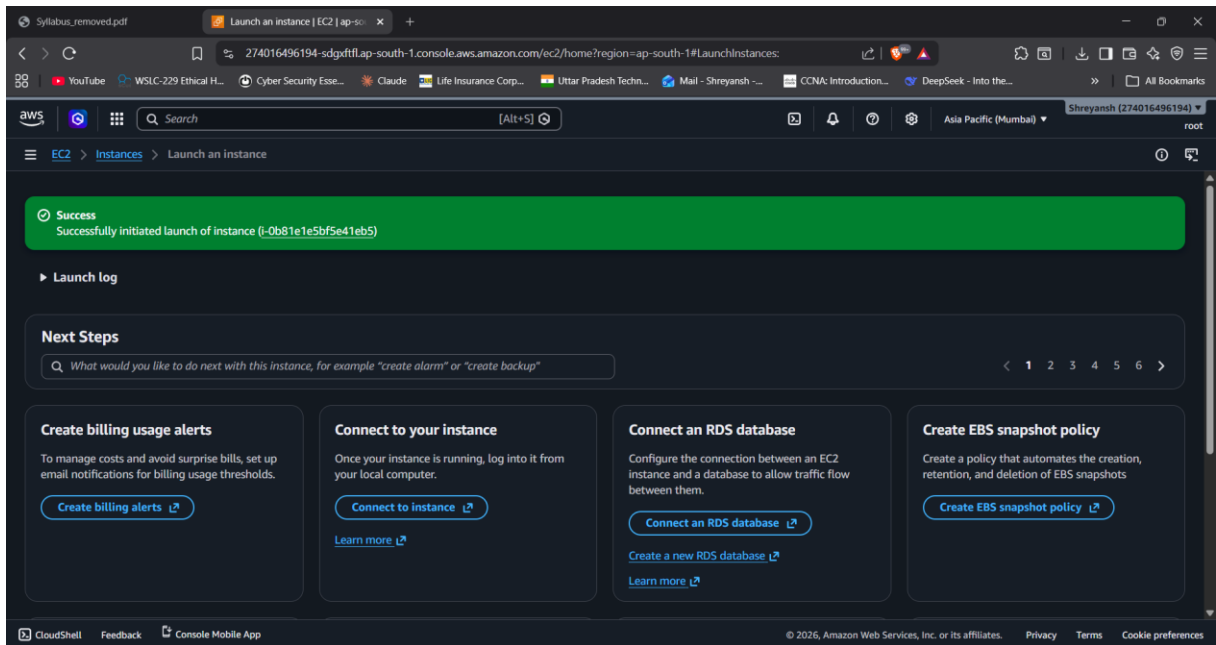
Step 6: Verify EC2 Instance Creation

The EC2 instance launch process completes successfully.



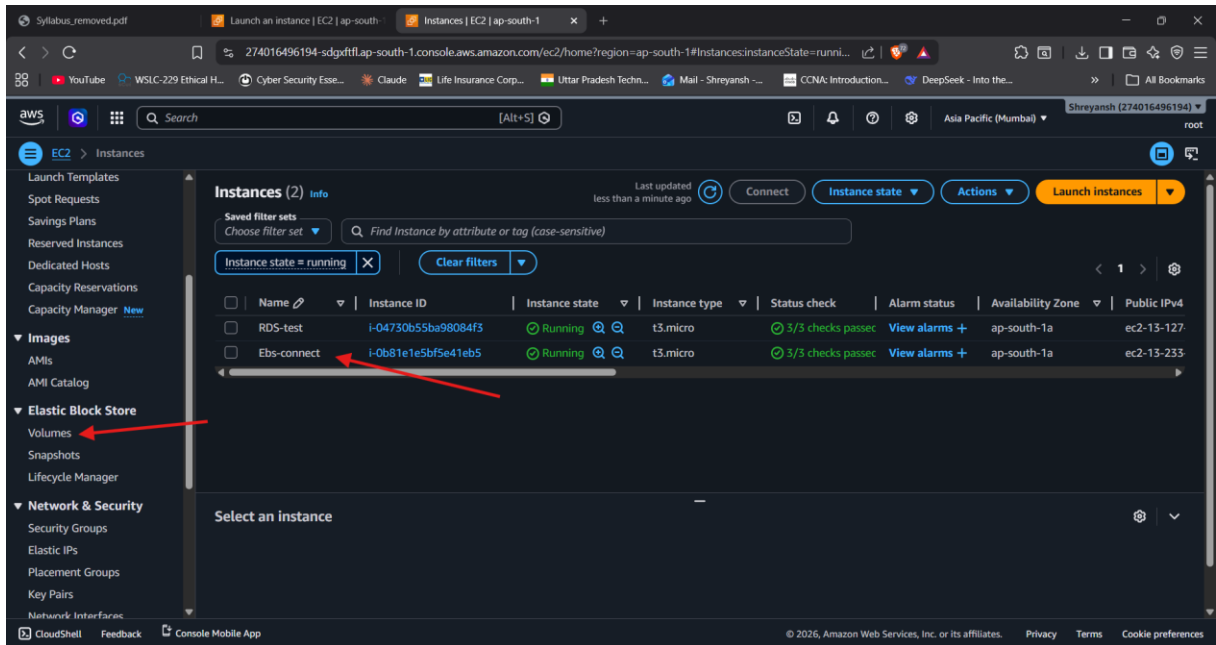
Step 7: View Running Instances

The running EC2 instances list confirms successful deployment.



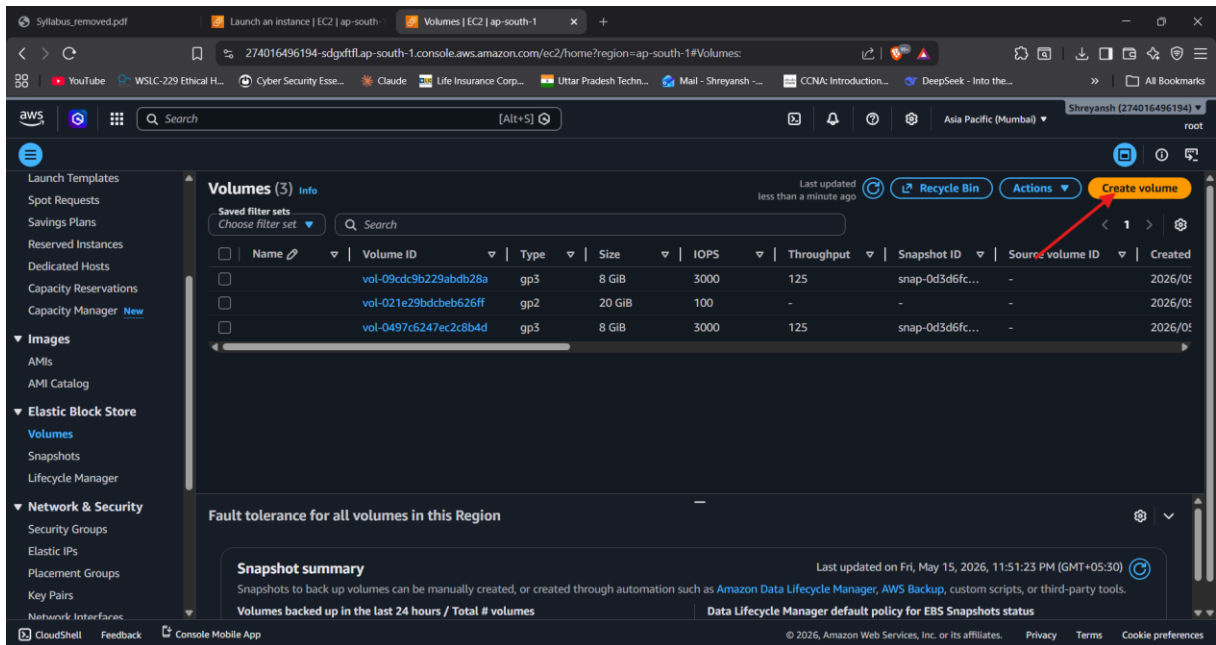
Step 8: Open EBS Volumes Section

The Elastic Block Store volumes dashboard is opened.



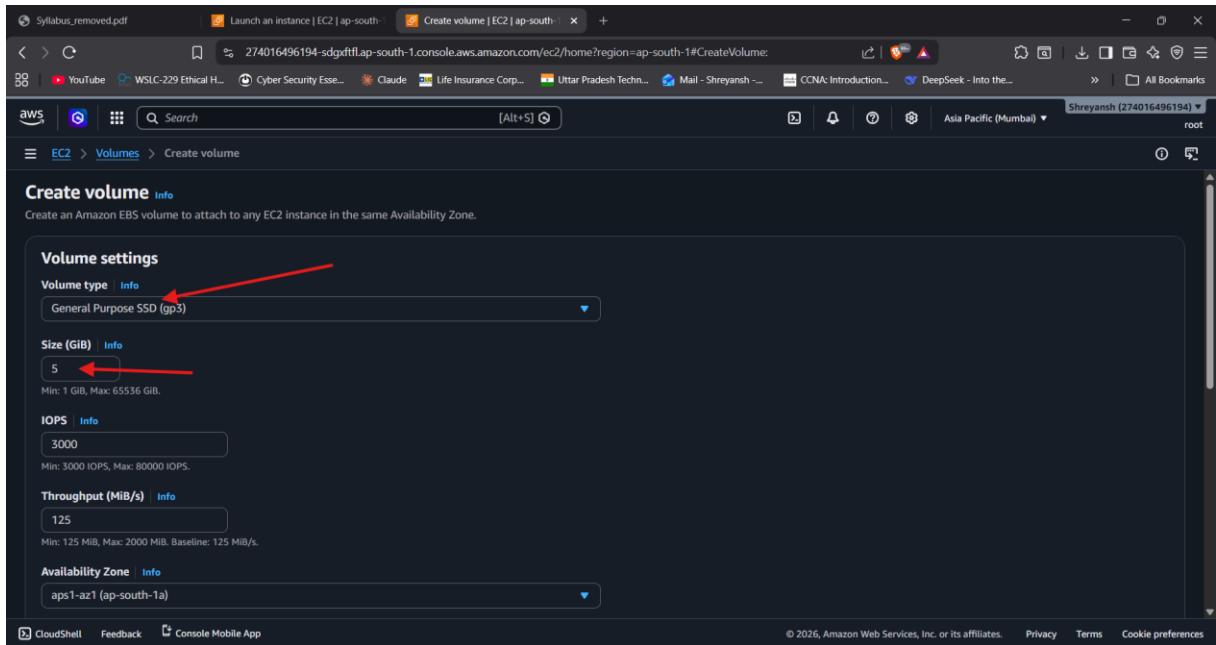
Step 9: Create a New EBS Volume

A new General Purpose SSD (gp3) EBS volume of 5 GiB is configured.



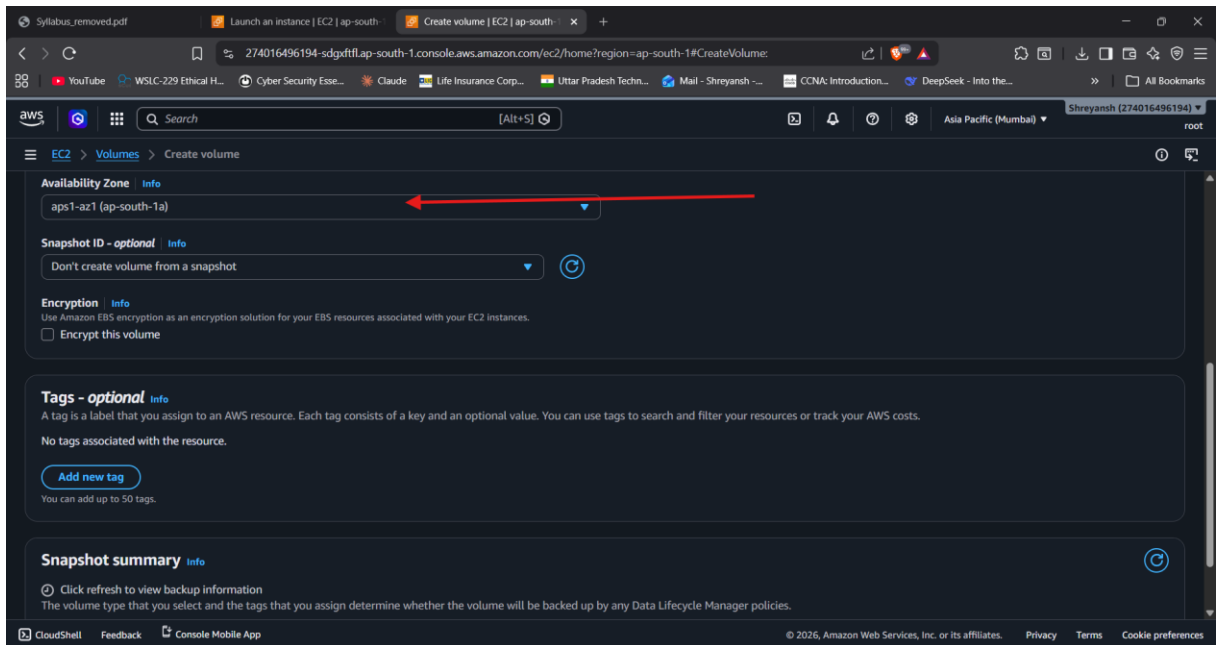
Step 10: Configure Availability Zone

The volume is configured in the same Availability Zone as the EC2 instance.



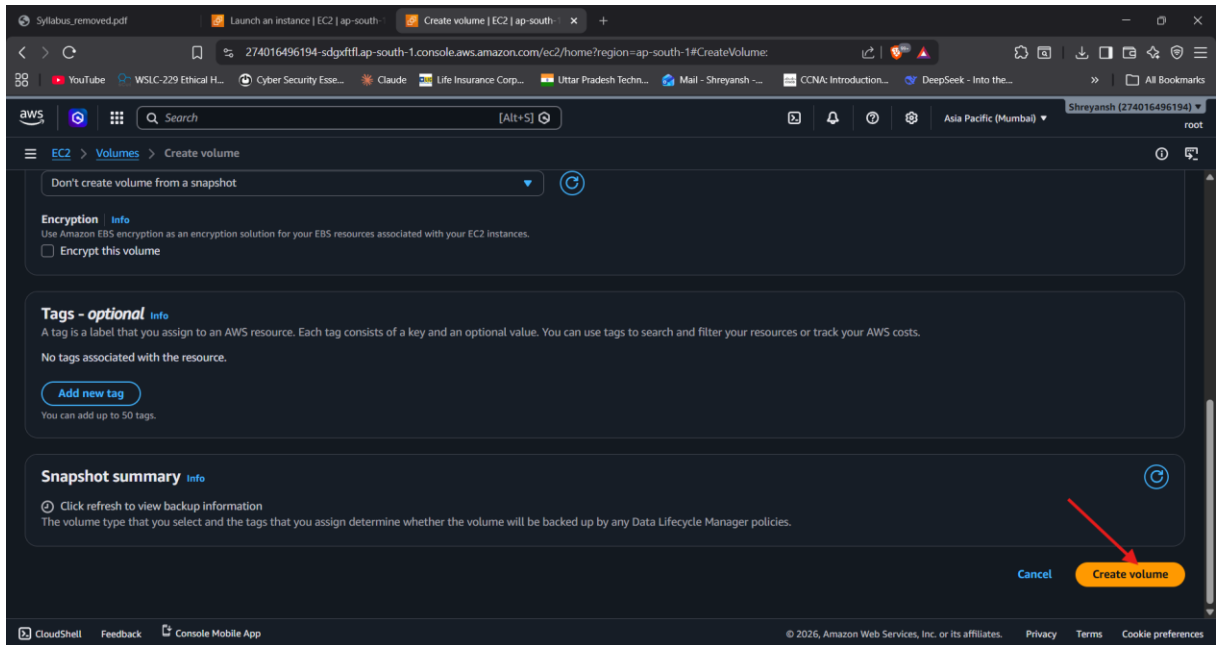
Step 11: Submit Volume Creation

The Create Volume action is submitted successfully.



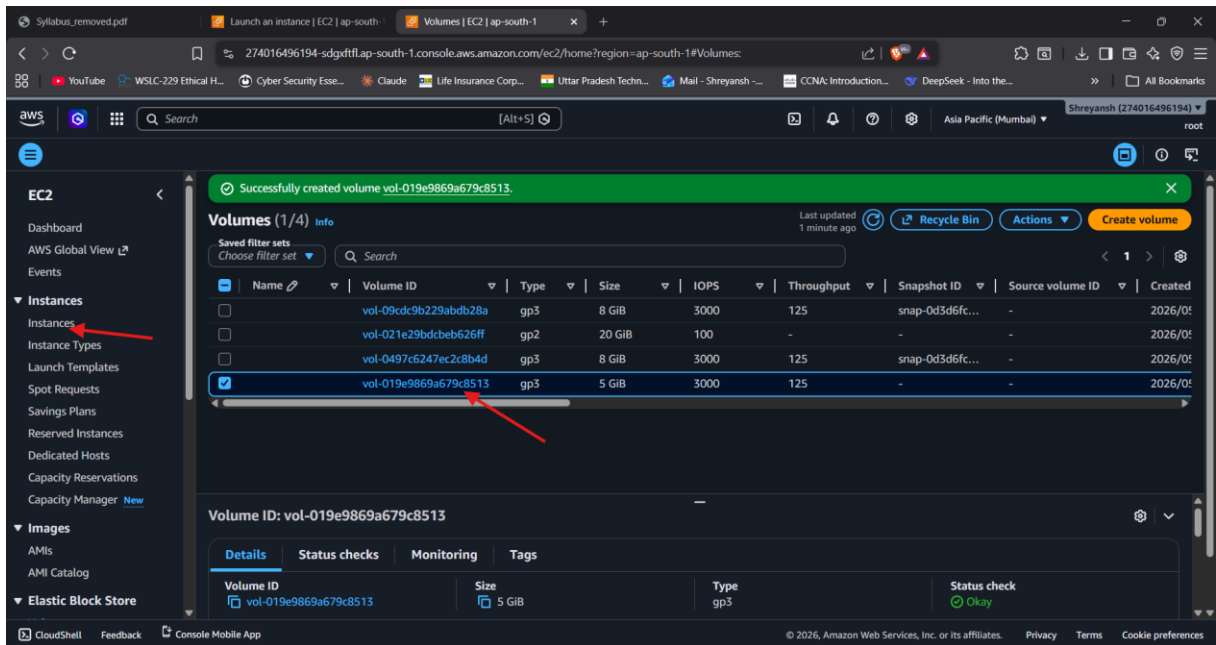
Step 12: Verify Created Volume

The newly created EBS volume appears in the Volumes dashboard.



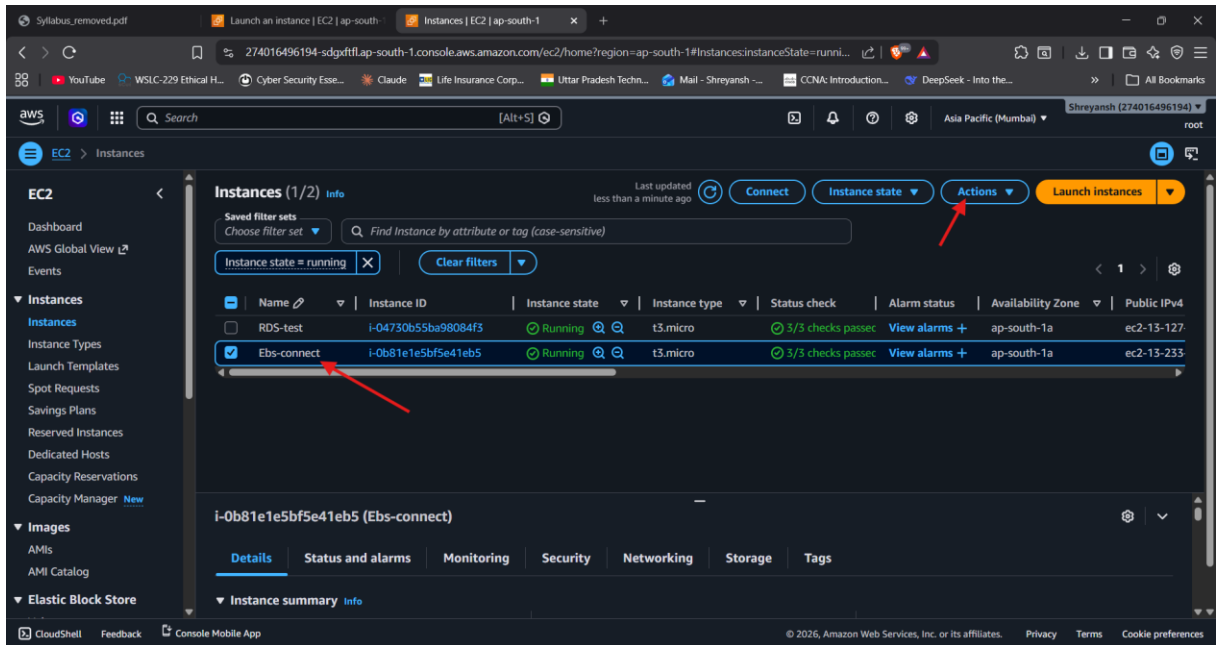
Step 13: Attach Volume to EC2

The Attach Volume option is selected from the EC2 storage menu.



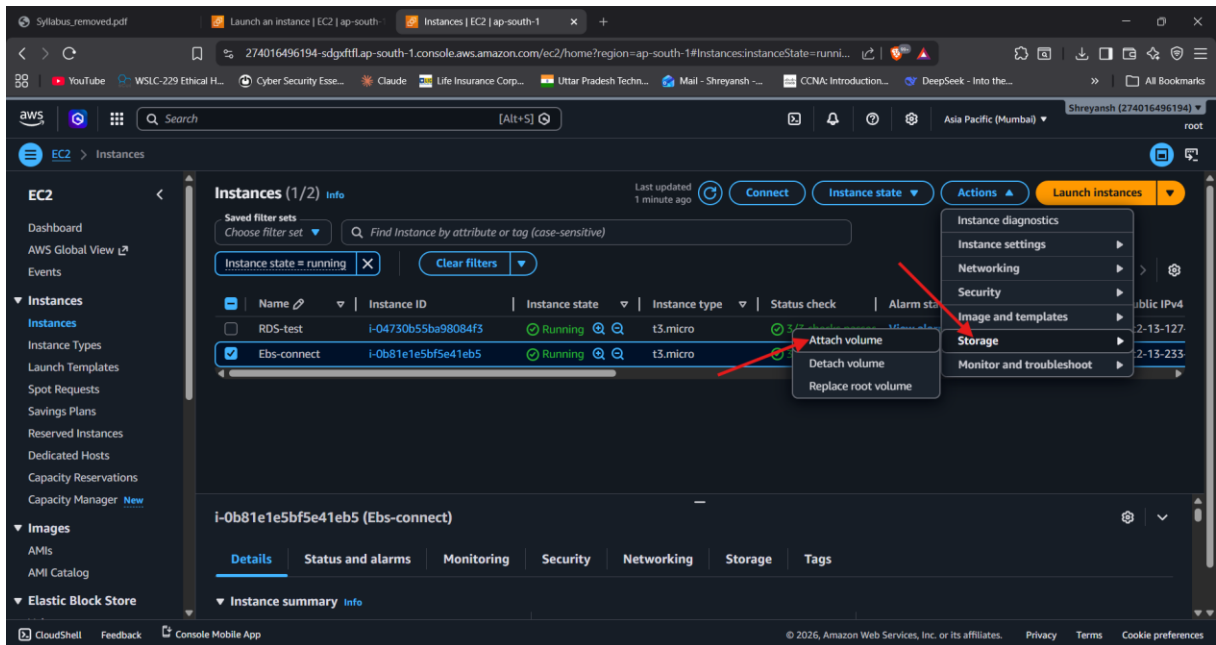
Step 14: Select Device Name and Attach

The EBS volume is attached using the Linux device name `/dev/sdb`.



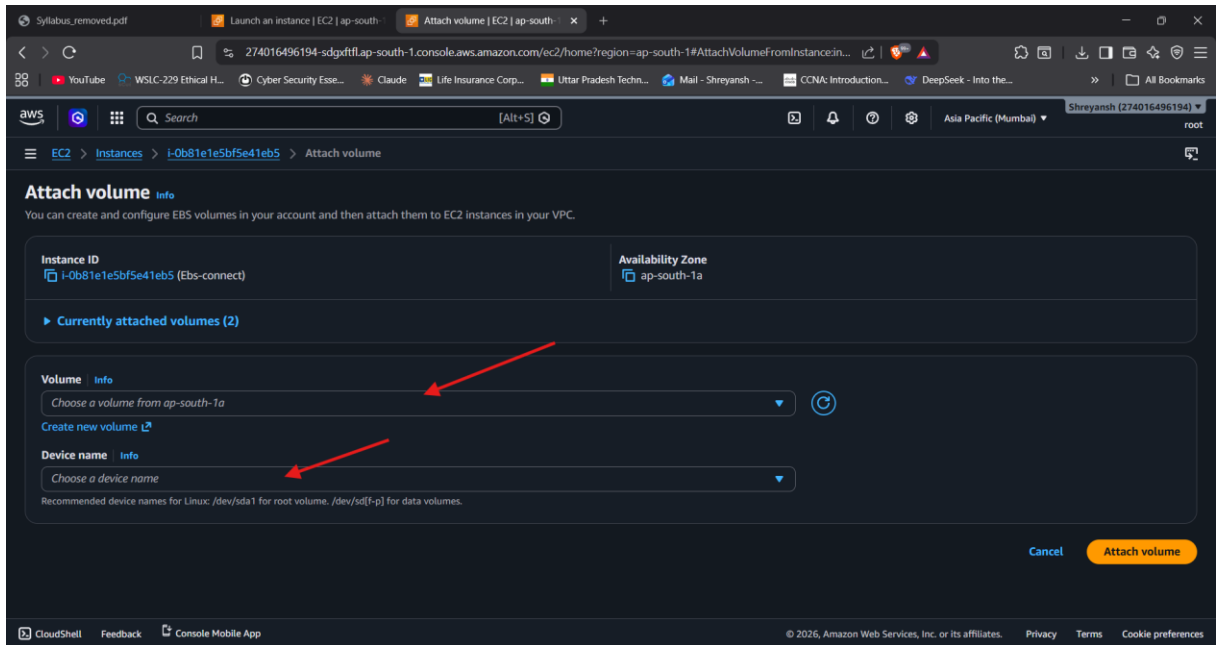
Step 15: Verify Attached Volume

The Storage tab confirms that the new EBS volume is attached and in-use.



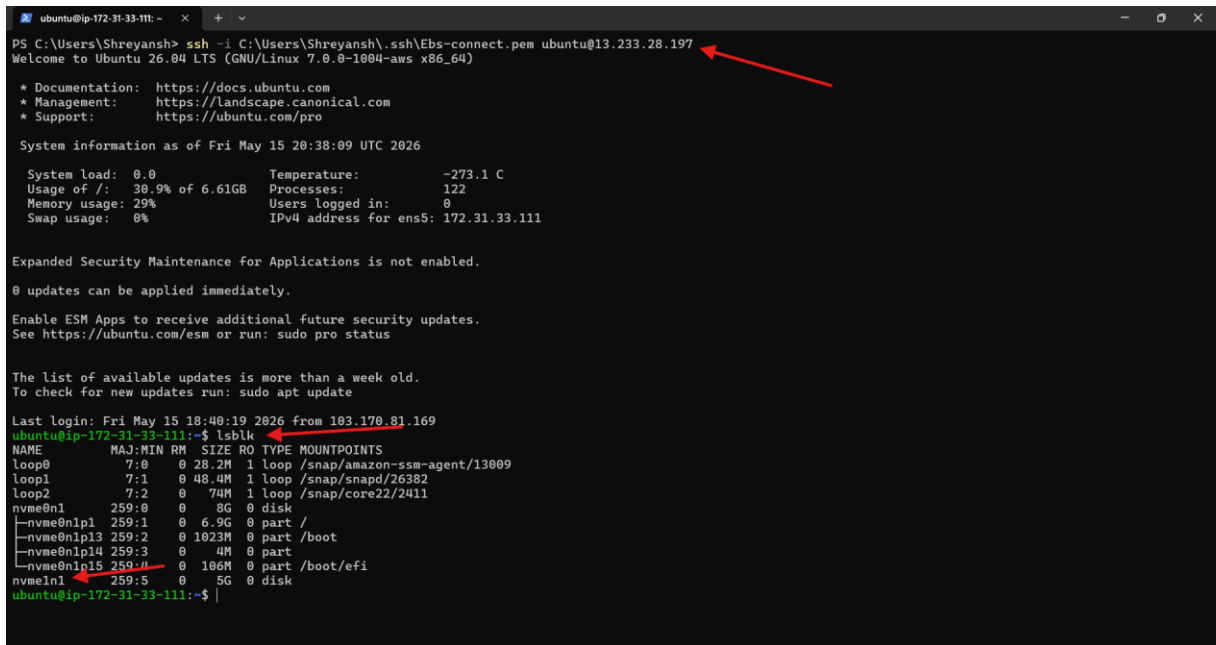
Step 16: Open Windows Terminal

Windows Terminal is opened for SSH connection to the EC2 instance.



Step 17: Connect to EC2 and Verify Disk

SSH login succeeds and the lsblk command shows the newly attached EBS volume.



Step 18: Format, Mount, and Verify EBS Volume

The EBS volume is formatted using ext4, mounted at /mnt/mydata, and verified successfully.

```
ubuntu@ip-172-31-33-111: ~$ lsblk
loop2      7:2    0   74M  1 loop  /snap/core22/2411
nvme0n1    259:0    0    8G  0 disk  /
├─nvme0n1p1 259:1    0   6.9G  0 part  /
├─nvme0n1p13 259:2    0  1023M  0 part  /boot
├─nvme0n1p14 259:3    0    4M  0 part  /
├─nvme0n1p15 259:4    0  106M  0 part  /boot/efi
└─nvme1n1   259:5    0    5G  0 disk  /
ubuntu@ip-172-31-33-111:~$ sudo mkfs -t ext4 /dev/nvme1n1
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: 91714a5d-f796-483c-b03e-599bfb5b80c3f
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

ubuntu@ip-172-31-33-111:~$ sudo mount /dev/nvme1n1 /mnt/mydata
ubuntu@ip-172-31-33-111:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.7G  2.1G  4.6G   31% /
tmpfs            455M    0  455M   0% /dev/shm
tmpfs           182M  912K  181M   1% /run
efivarfs        128K   3.1K   128K   3% /sys/firmware/efi/efivars
none            1.0M    0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           455M    0  455M   0% /tmp
none            1.0M    0  1.0M   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13 989M  96M  826M  11% /boot
/dev/nvme0n1p15 105M  6.3M   99M   7% /boot/efi
none            1.0M    0  1.0M   0% /run/credentials/systemd-networkd.service
none            1.0M    0  1.0M   0% /run/credentials/getty@tty1.service
none            1.0M    0  1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs           91M   8.0K   91M   1% /run/user/1000
/dev/nvme1n1     4.9G  1.3M  4.6G   1% /mnt/mydata
ubuntu@ip-172-31-33-111:~$ echo "Hello from EBS!" | sudo tee /mnt/mydata/test.txt
Hello from EBS!
ubuntu@ip-172-31-33-111:~$ cat /mnt/mydata/test.txt
Hello from EBS!
ubuntu@ip-172-31-33-111:~$
```

Linux Commands Used

- lsblk
- sudo mkfs -t ext4 /dev/nvme1n1
- sudo mount /dev/nvme1n1 /mnt/mydata
- df -h
- echo "Hello from EBS!" | sudo tee /mnt/mydata/test.txt
- cat /mnt/mydata/test.txt

Final Result

The AWS EC2 and EBS practical was completed successfully. The EC2 Ubuntu instance was launched, the EBS volume was created and attached, and Linux commands verified successful mounting and storage access. The test file created inside the mounted directory confirmed that the EBS volume was functioning correctly.

Conclusion

This experiment demonstrated the practical implementation of Amazon EC2 and Elastic Block Store (EBS) in AWS Cloud. The activity covered instance deployment, volume management, SSH connectivity, Linux filesystem formatting, and storage verification. The lab also highlighted the importance of security groups, key pairs, and device mounting in real-world cloud administration tasks.